

DeepDTA-iBAM: Cross-Attention Interaction Mapping for Affinity Prediction, Ligand Retrieval, and Analog Generation

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ABSTRACT Drug–target affinity models can support screening through affinity scores, interaction maps, and candidate ranking. We present DeepDTA-iBAM, a 41.5 M-parameter architecture that integrates these outputs with protein-conditioned analog generation. A graph encoder represents ligands, cached protein language-model embeddings represent targets, and bidirectional cross-attention couples atom and residue tokens for affinity regression, interaction maps, and retrieval. Analog generation shares the protein adapter and checkpoint. Retaining the maps adds a measured mean overhead of 1.5% to CPU inference time. On a custom random split of the KIBA kinase benchmark, the model achieves a concordance index of 0.862 and root-mean-square error (RMSE) of 0.452. At the conventional activity threshold, the area under the receiver operating characteristic curve (AUROC) is 0.924 and the area under the precision-recall curve is 0.832, against a positive base rate of 0.214. Retraining ablations across three seeds do not isolate a dominant component. Zeroing the trained cross-attention gates raises RMSE by 33% and reduces concordance by 0.080. On the docked subset of an assay-defined epidermal growth factor receptor panel, the model exceeds AutoDock Vina by 0.188 AUROC but trails fingerprint similarity by 0.156. We identify two benchmark constructions that prevent valid interpretation and introduce three reporting checks to detect them. These results support integrated prediction and analysis while clarifying the limits of interaction-map interpretation and screening performance.

INDEX TERMS Attention mechanisms, benchmark testing, biomedical computing, drug discovery, machine learning, molecular biophysics, neural network architecture, performance evaluation.

I. INTRODUCTION

PREDICTING how tightly a small molecule binds a protein target is a core problem in computational drug discovery, and shared datasets such as KIBA have supported comparisons among successive model generations, from similarity-based methods through convolutional, graph-based, and transformer-inspired architectures [1]–[13].

A single predicted number rarely supports a decision. A project team wants to know which part of the pocket the prediction rests on, which compounds to order, and what to make next [14]. Meeting those needs with separate models gives no guarantee that the explanation corresponds to the prediction it is meant to explain; meeting them with one model requires an architecture whose internal representations are shared across tasks by construction.

The original DeepDTA architecture encodes ligand and protein through parallel convolutional stacks and concatenates the resulting vectors [4]. Nothing in that design represents which ligand atom interacts with which residue, so the

network emits an affinity scalar and nothing else. Later work added graph ligand representations [8], [10], [15], transformer encoders with substructure interaction maps [9], and learned protein representations with joint attention [6]. These approaches motivate integrating predictive and interpretive outputs within a shared model.

A. APPROACH

DeepDTA-iBAM introduces a bidirectional cross-attention interface between ligand atom tokens and protein residue tokens. The interface is not an add-on for interpretability: it is the fusion mechanism through which the affinity head receives information, so the atom-by-residue coupling it computes contributes directly to the prediction. The fused representation also supports latent retrieval; analog proposal shares the protein adapter and checkpoint.

Building such a system raises an evaluation problem we treat as part of the contribution. Affinity prediction has widely used datasets and evaluation metrics, whereas benchmarks for

the secondary capabilities are often assembled by the authors of each study. Building those panels ourselves lets us state exactly how each positive class was defined, and two reveal metrics that are undefined or confounded by the selection rule.

B. CONTRIBUTIONS

- 1) **A unified multi-task architecture.** A bidirectional cross-attention interface couples a graph ligand encoder to cached protein language-model embeddings, so one checkpoint serves affinity regression, interaction-map extraction, and retrieval through fusion, with analog generation conditioned on the shared protein adapter (Section II).
- 2) **The cost of collecting interaction maps.** We report parameter counts, CPU latency, throughput, and how these vary with target length. Collecting the already-computed maps adds a measured mean overhead of 1.5%. Target length is associated with latency and affects batch memory (Section IV-C).
- 3) **An inference-time ablation of the fusion pathway.** Retraining ablations at three seeds cannot single out any one component. Zeroing the learned cross-attention gates in the trained checkpoint instead costs 0.080 concordance index and raises root-mean-square error (RMSE) by a third, demonstrating dependence on this pathway in the trained model (Section IV-B).
- 4) **A direct test of what the interaction maps encode.** We hold the epidermal growth factor receptor (EGFR) fixed and vary the ligand across 3,300 assay-defined actives and property-matched decoys, then split the attention variance into ligand-shared and ligand-specific parts. A linear probe then asks how much predicted affinity and active-decoy labels the maps carry, including for decoys without reference poses (Section IV-D).
- 5) **Three reporting checks for benchmark validity.** Checks of the base rate, chance-referenced overlap, and baseline independence identify two benchmark failures with little additional computation. They distinguish a repairable benchmark from one that requires rebuilding (Section V).

Prior studies of benchmark bias inform this contribution. Benchmarks may reward memorization when positives are congeneric [16], [17], when uncontrolled property differences let trivial features carry the score [18], or when hidden structure lets a deep network score well while learning nothing about protein–ligand interaction [19]. Property-matched decoys are the standard response [20], [21]. Our two constructions differ in kind: one makes the area under the receiver operating characteristic curve (AUROC) undefined, while the other makes a baseline score inseparable from the panel’s selection rule.

II. SYSTEM ARCHITECTURE

DeepDTA-iBAM couples ligand and protein representations through fusion, with a generation branch conditioned on the protein adapter (Fig. 1).

A. LIGAND ENCODER

Molecular structures in simplified molecular-input line-entry system (SMILES) notation are converted to atom–bond graphs with RDKit and encoded by a multi-head graph attention network [22] with edge-aware masking, using 6 layers, 8 heads, and a hidden width of 512. Edge features carry bond order and ring membership, and each layer applies layer normalization and a residual connection. Attention pooling produces a graph-level vector alongside the per-atom token sequence, which is what the fusion stage consumes.

B. PROTEIN ENCODER

Residues are represented by cached embeddings from ESM-C 600M [23], a Cambrian-series protein language model that differs from ESM-2 [24]. Embeddings are computed once per unique sequence and cached to disk. After cache construction, the language model need not run during affinity training or inference for cached targets. A protein adapter projects them to the shared width, applies a depthwise convolution over sequence position to capture local motifs, and passes the result through self-attention blocks. Caching is what makes the system practical: the protein encoder is the most expensive component, and screening against one target reuses a single cached embedding for every compound.

C. BIDIRECTIONAL CROSS-ATTENTION FUSION

The fusion stage distinguishes this architecture. Atom tokens $A \in \mathbb{R}^{n_a \times d}$ and residue tokens $R \in \mathbb{R}^{n_r \times d}$ enter three stacked blocks. Here, n_a and n_r are atom and residue counts, d is the token width, and MHA denotes multi-head attention. Each block computes attention in both directions,

$$\tilde{A} = \text{MHA}(A, R, R), \quad \tilde{R} = \text{MHA}(R, A, A), \quad (1)$$

with padding masked on the key side, and updates each stream through a learned scalar gate,

$$A \leftarrow A + g_a \tilde{A}, \quad R \leftarrow R + g_r \tilde{R}, \quad (2)$$

followed by a position-wise feed-forward block. The atom-to-residue attention weights form an $n_a \times n_r$ matrix, the interpretable binding attention map (iBAM). Averaging it over heads, over the three blocks, and then over a ligand’s real atoms gives a per-residue interaction profile.

Two properties follow from this fusion design. The map is the coupling used to construct the affinity head’s input. It requires no separate attribution pass, because the weights arise within the forward pass. Section IV-C measures what retaining them costs and Section IV-D tests what they encode, while the gates in (2) being learned scalars gives a clean way to ablate the pathway at inference (Section IV-B).

D. AFFINITY AND GENERATIVE HEADS

The affinity head consumes the pooled graph vector, protein vector, and fused state through a residual multilayer perceptron. Training combines a Huber loss ($\delta = 1.0$), a pairwise ranking term weighted 0.4 after a three-epoch warmup, and a diffusion

auxiliary ramping to 0.02 between epochs 2 and 6. A target-conditioned denoising diffusion head [25] operates on the ligand atom-feature tensor and is reused at inference to propose analogs around a seed structure. Its conditioning vector is pooled from the protein adapter before fusion, and generation bypasses the cross-attention blocks.

III. EXPERIMENTAL SETUP

A. TRAINING DATA AND CHECKPOINT

The model was trained on a random 90/5/5 split of KIBA [1], generated with seed 4221 and termed the standard split here (106,428 training, 5,913 validation, and 5,913 test pairs), on one GPU. Evaluation uses the released checkpoint labeled epoch 47 in both its metadata and optimizer state. A separate training summary ends at epoch 44 and does not document selection of that checkpoint. Every analysis uses that single checkpoint, so differences between experiments reflect evaluation design rather than retraining. The one exception is the archived component ablation, which aggregates three re-training seeds and so does not reproduce the single-checkpoint values exactly.

This chemically permissive split differs from published benchmark folds. We therefore report overlap diagnostics and also evaluate a scaffold split.

B. STRUCTURAL PANEL

Structural analyses use five kinase co-crystal complexes: 6YOJ (FAK1, ligand P4N), 4WKQ (EGFR, IRE), 2HYY (ABL1, STI), 1KE6 (CDK2, LS2), and 4RJ3 (CDK2, 3QS), with contacts defined by a 4.5 Å heavy-atom cutoff throughout. The panel spans four kinase targets, with two distinct CDK2 complexes.

C. RETRIEVAL PANELS

Three retrieval panels differ in exactly the respect that determines what each can measure: how the positive class was defined. Because our claim is that panel construction determines validity, each rule is stated in full and all are deterministic in the released code.

1) EGFR family panel (similarity-defined)

Activity records for ChEMBL target CHEMBL203 were retrieved with assay type B, standard type in $\{K_i, K_d, IC_{50}\}$, and standard relation =, requiring a non-null pChEMBL value, the negative base-10 logarithm of the reported molar potency, and a canonical SMILES string. Relation operator and measurement type are separate filters and should not be collapsed into one qualifier: = excludes censored values such as > or <, while the type filter selects which readouts are admitted. K_i and K_d describe inhibition and dissociation, respectively, whereas IC_{50} is a condition-dependent half-maximal inhibition readout.

Where a pair had several qualifying records, the panel retained the highest pChEMBL value, ties broken by the lowest nanomolar value, irrespective of measurement type. Prior work instead imposes a type priority, preferring K_d over K_i and K_i

over IC_{50} [26]; Section III-D reports a sensitivity analysis under that priority.

Retained molecules were filtered to $pChEMBL \geq 7.0$ and Morgan extended-connectivity fingerprint (ECFP4, radius 2, 2048 bits) [27] Tanimoto similarity ≥ 0.40 to gefitinib (CHEMBL939) or erlotinib (CHEMBL553), leaving 347 molecules. Tanimoto similarity is the fraction of substructure features two molecules share, so it runs from 0 for no overlap to 1 for identical fingerprints. Those two are the canonical first-generation EGFR inhibitors and 0.40 a conventional relatedness cutoff; neither is load-bearing, because the defect below follows from the existence of the similarity filter, not its value.

Six anchors were selected deterministically, not by random seed: the gefitinib and erlotinib records first, then molecules added greedily, each time maximizing

$$\min_{j \in S} (1 - T(i, j)) + 0.03 p_i, \quad (3)$$

where T is Tanimoto similarity to the selected set S and p_i the pChEMBL value. The remaining 341 became holdout positives.

Decoys came from a local property-filtered lead-like ZINC tranche archive [28]. The first 5,000 lines of each tranche file were scanned and a per-tranche quota of $\lceil 2000/n_{\text{tranches}} \rceil$ was sampled without replacement under a fixed seed. Any SMILES already in the anchor or holdout sets was excluded, with deduplication on canonical SMILES throughout. This gives 2,000 decoys and a candidate set of 2,341.

2) Histamine H1 panel (assay-defined, out of domain)

Twenty H1 receptor ligands were selected from one ChEMBL page of 1,000 assay-type B records, with no standard-type or relation filter. Selection required $pChEMBL \geq 6.0$ or potency $\leq 1,000$ nM and prioritized a fixed list of named antihistamines, followed by other named potent candidates to reach 20 unique SMILES. Candidates were ranked by descending pChEMBL and ascending nanomolar potency. The released set contains 13 curated-name entries and 7 fallback entries; clinical phase was recorded when available but was not an eligibility criterion. Five replicate libraries each contained the same 20 actives and 9,980 lead-like compounds drawn from the ZINC bank at five consecutive fixed seeds, giving 10,000 molecules and a positive base rate of 0.002 per library. No similarity filter was applied at any stage.

3) In-domain EGFR panel (assay-defined)

The Directory of Useful Decoys, Enhanced (DUD-E), supplies EGFR actives defined by ChEMBL activity and property-matched decoys [20]. We sampled 300 actives against 3,000 decoys under a fixed seed.

Assay outcome defines the positive class in the second and third panels, while the first additionally imposes a similarity cutoff. Fingerprint results on the third panel remain subject to DUD-E's similarity-based decoy selection.

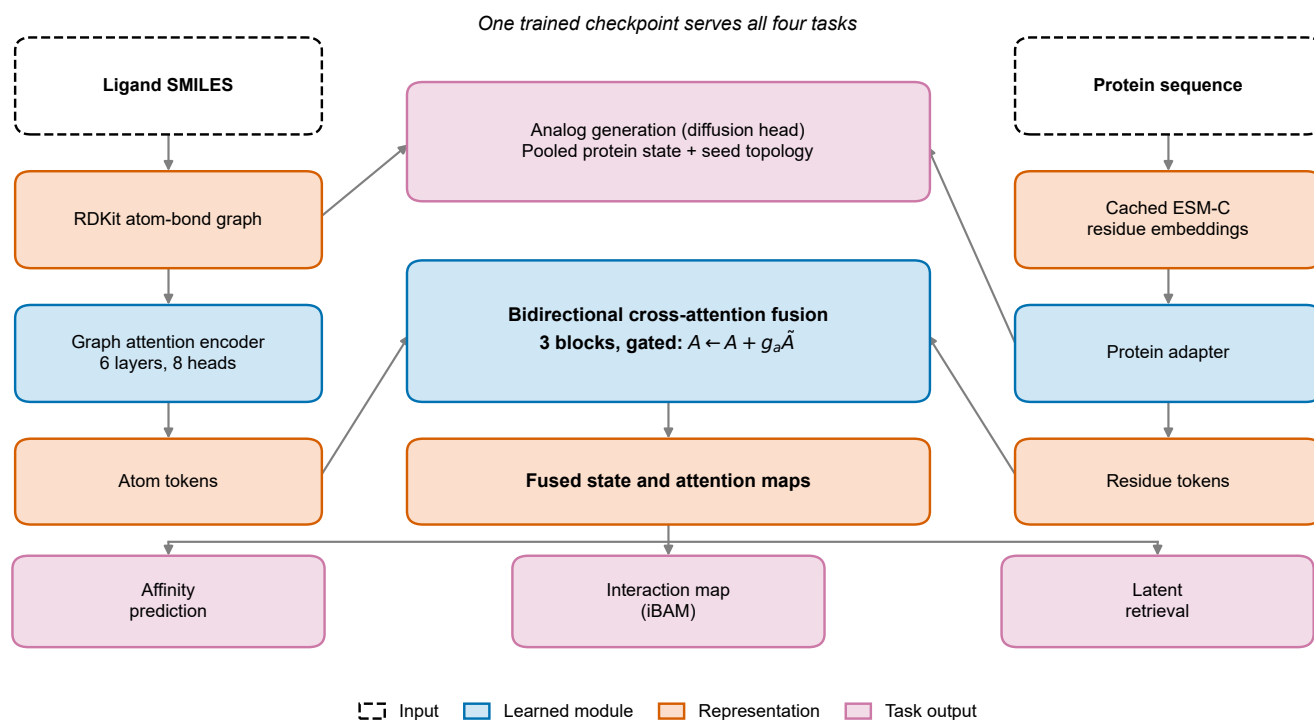


FIGURE 1. DeepDTA-iBAM architecture. A graph attention encoder turns the ligand into atom tokens, while cached ESM-C embeddings compressed by a protein adapter give residue tokens. Three gated bidirectional cross-attention blocks fuse the two streams for affinity prediction, interaction maps, and latent retrieval. Generation instead uses the pooled protein-adaptor vector and the seed graph. The affinity head also receives pooled pre-fusion ligand and protein vectors (omitted here for clarity). Node class is encoded by fill and border colors, with dashed borders for inputs, as given in the legend. The atom-by-residue attention matrix reported throughout this paper is the coupling computed inside the fusion stage, so it is a byproduct of the same forward pass that produces the affinity prediction rather than a separate attribution step.

D. SENSITIVITY TO THE ACTIVITY CONSOLIDATION RULE

To assess sensitivity to consolidation, we analyzed 19,527 cached qualifying ChEMBL203 records, covering 10,583 distinct pairs. The mix is heavily skewed: 18,119 records are IC_{50} , 875 K_d , and 533 K_i , and only 180 pairs, 1.7%, carry more than one type, so for the large majority the two rules cannot disagree.

Re-deriving every pair under the $K_d > K_i > IC_{50}$ priority changes which record is retained for 64 pairs and the recorded potency for 55, with a maximum shift of 4.84 log units where a pair carries both a weak binding constant and a potent functional readout. At the panel’s pChEMBL ≥ 7.0 gate the retained set moves from 5,942 parent IDs to 5,928, with 14 dropping out and none entering, a Jaccard index of 0.998. Applying the similarity cutoff to this cache gives 756 versus 752 candidates. This larger extraction does not reproduce the original 347-candidate panel or re-estimate its retrieval metrics.

E. EVALUATION PROTOCOL

The main KIBA evaluation used 2,000 bootstrap resamples at a fixed seed; the earlier evaluation retained in the supplement used 600. Literature benchmark values are primary-source results under their published splitting protocols [4], [5], [8], [10], [15], not rerun here, so that comparison is contextual

rather than a leaderboard.

Rankers scored on a shared candidate set have correlated errors, so comparing marginal confidence intervals is not a valid test of their difference; retrieval comparisons use a paired bootstrap with 10,000 resamples at a fixed seed. The inference-time ablation uses 2,000 paired resamples.

For retrieval we report the area under the receiver operating characteristic curve (AUROC), the area under the precision-recall curve (AUPRC), BEDROC20, enrichment factors, and recovery at ranked fractions, each beside the positive base rate.

How the first three behave matters for the rest of the paper. AUROC is the probability that a randomly chosen active outranks a randomly chosen decoy, and it is independent of prevalence for fixed class-conditional score distributions. AUPRC depends on prevalence: the positive fraction is its random-ranking reference, not a lower bound. BEDROC20 is Boltzmann-enhanced discrimination of ROC with $\alpha = 20$, emphasizing early retrieval, approximately the first 8% of the ranked list.

For residue-level contact AUROC we compute an analytic standard error per complex [29] and pool by inverse-variance weighting, reporting fixed-effect and random-effects estimates [30]. Derivations are provided in the supplement.

F. DOCKING REFERENCE

AutoDock Vina 1.2.7 [31], [32] provides an external structure-based reference. The receptor is chain A of 4WKQ with waters and cofactors removed, converted to PDBQT with Open Babel at pH 7.4 with Gasteiger charges, and the search box is centered on the crystallographic gefitinib with 8 Å padding. Ligands are embedded with RDKit ETKDGv3 at a fixed seed, optimized with MMFF, and converted with Meeko. Docking used exhaustiveness 8 and a fixed seed, and scores are negated before computing retrieval metrics because Vina reports an estimated binding free energy. Using Open Babel rather than the AutoDock tooling bounds the comparison.

Docking all 3,300 candidates requires substantial CPU computation, so the panel was subsampled. Because retrieval precision is governed by the smaller class and the panel holds one active per ten decoys, we docked every active and capped the decoys, drawing them from an order shuffled under a fixed seed. AUROC is prevalence-independent, but its bootstrap uncertainty depends on class counts; AUPRC and enrichment factors are reported at the realized prevalence.

All three rankers are evaluated on exactly the docked set. That matters for the fingerprint baseline, whose released scores are leave-one-out similarities against all 300 panel actives: scoring a subsample against that full reference set would hand it evidence the subsample does not contain. We recompute it within the docked set, which here coincides with the full-panel definition since every active was docked.

G. REPRODUCIBILITY

All analysis code and derived data are released (<https://github.com/kevinmsong/DeepDTA-iBAM>). Every statistic reported here is supported by scripts and archived outputs in `analysis/` and `results/`, and intermediate artifacts are released so downstream statistics can be rechecked without rerunning inference. Every random choice described below, from data splits to bootstrap resampling to decoy sampling, is seeded; the seed values are set in those scripts rather than repeated in the text.

A checkpoint audit records file hashes, verifies the epoch-47 optimizer state, and compares fresh CPU predictions with archived outputs. The incomplete training history does not establish the checkpoint's original selection decision. Supplementary audits also recheck atom ordering, candidate curation, and probe validation while preserving the original artifacts.

The release includes a minimal CPU inference endpoint. Original weights and protein caches are not distributed; trained inference requires those assets. An explicit demonstration with random weights checks tensor flow only.

Inference scripts adjust batch size using the longest target and available memory, allowing screening runs to adapt to the host. The checkpoint occupies 166 MB and the combined graph and protein caches 509 MB, including 385 MB of protein embeddings. CPU inference was demonstrated on a workstation; training used a GPU.

TABLE 1. Predictive accuracy on the KIBA standard-split test partition.

Metric	Value	95% CI
<i>Regression on the KIBA score</i>		
RMSE	0.4517	[0.4369, 0.4673]
MAE	0.2950	—
Pearson r	0.8741	—
Spearman ρ	0.8484	[0.8373, 0.8588]
R^2	0.7049	—
Concordance index	0.8625	[0.8568, 0.8681]
<i>Classification at KIBA ≥ 12.1</i>		
Positive base rate	0.2143	—
AUROC	0.9240	[0.9151, 0.9334]
AUPRC	0.8316	[0.8139, 0.8503]
Balanced accuracy	0.8587	—
Matthews correlation	0.6942	—

$n = 5,913$ pairs. Lower is better for RMSE and MAE (KIBA-score units); higher is better for the other performance metrics. The base rate describes class composition. Intervals are 95% percentile bootstrap over 2,000 resamples; CI denotes confidence interval. Dashes indicate intervals not reported.

The concordance index counts prediction ties as half a concordance and excludes pairs with equal measured values, the convention the comparison methods use.

IV. RESULTS

A. PREDICTIVE ACCURACY

On the KIBA test partition the model reaches a concordance index of 0.8625 (95% bootstrap interval 0.8568 to 0.8681) and RMSE 0.4517 (0.4369 to 0.4673). The concordance index is the share of pairs the model orders correctly, so 0.5 is chance and 1.0 is a perfect ranking. It is the standard measure here because ranking, not the absolute value, is what a screening decision uses. Binarizing at the conventional activity threshold of 12.1 gives AUROC 0.9240 and AUPRC 0.8316, which is 3.88 times the positive base rate of 0.2143 (Table 1).

For context, the original DeepDTA baseline reports 0.863; GraphDTA, DGraphDTA, and MGraphDTA report 0.891 to 0.904 under their published splitting protocols [4], [8], [10], [15]; because those methods were not rerun on our split, these values do not establish performance parity or a ranking. The architectural contribution is the shared multi-task interface; its accuracy tradeoff requires a controlled comparison.

Two qualifications matter for deployment. Residuals have a mean bias of -0.0095 with limits of agreement from -0.895 to 0.876 , wide enough that a single predicted value should not drive a commitment to one compound. And all 1,587 compounds and 219 targets in the test partition also occur in training, so this split evaluates unseen pairs of known compounds and targets, without testing unseen compounds or targets. Under a scaffold split, the same configuration reaches a concordance index of 0.7615.

B. COMPONENT ABLATION

Two ablations are reported, because they answer different questions.

TABLE 2. Inference-time ablation of the cross-attention pathway, on 1,200 held-out KIBA pairs.

Condition	Concordance	RMSE	Pearson r
Intact	0.8613	0.4475	0.8760
Gates halved	0.8508	0.4521	0.8641
Gates zeroed	0.7809	0.5948	0.7355

1) Retraining ablation: no component is resolved

The archived ablation retrains each variant from scratch with three seeds (Table 3). Removing cross-attention, the ranking loss, or the diffusion head changes standard-split concordance index by at most 0.007 and scaffold-split by at most 0.009, with every contrast returning $p \geq 0.42$ and $|d| \leq 0.74$ at $n = 3$, and on the standard split every ablation has a nominally higher mean than the full configuration.

Estimates from three seeds per variant are too imprecise to attribute predictive accuracy to an individual component. The multi-task results establish that the components provide the additional capabilities; they do not establish an improvement in regression performance.

2) Inference-time ablation: what the model relies on

A different and answerable question is how much the model *as trained* depends on the cross-attention pathway. Setting the learned gates in (2) to zero removes cross-modal flow while leaving every other weight exactly as trained, so the degradation is attributable to the pathway rather than to a different training run reaching a different optimum.

Both gates were initialized at 0.1. After training, gates updating atom tokens settled at 0.045, 0.040, and 0.061 across the three blocks, while gates updating residue tokens grew to 0.188, 0.183, and 0.197. Gate magnitudes alone do not quantify directional importance because the attention updates can differ in scale.

Zeroing them degrades the model substantially (Table 2). On 1,200 held-out KIBA pairs the intact checkpoint reaches concordance index 0.8613 and RMSE 0.4475, consistent with the full-test estimates. With the gates off, concordance falls to 0.7809 and RMSE rises to 0.5948. The bootstrap mean intact-minus-zeroed RMSE difference is -0.148 KIBA units (95% CI -0.190 to -0.107), corresponding to a 33% increase in RMSE, with concordance index down 0.080. Halving the gates reduces concordance index by 0.0105, with the interval for the RMSE difference spanning zero.

So the trained model does depend on the fusion pathway even though the retraining ablation resolved nothing. These are not contradictory: a retrained model could compensate through its remaining capacity, whereas this one never had that opportunity. The inference-time result measures sensitivity to removing the pathway in this checkpoint. Resampling individual pairs does not capture training variability or dependence among pairs sharing compounds or targets.

TABLE 3. Retraining ablation: concordance index across three independently retrained seeds per variant.

Split	Variant	Concordance (mean $\pm$ SD)	p
Standard	Full model	0.8562 ± 0.0098	–
	No cross-attention	0.8596 ± 0.0055	0.63
	No ranking loss	0.8634 ± 0.0096	0.42
	No diffusion head	0.8582 ± 0.0057	0.78
	Backbone only	0.8633 ± 0.0113	0.46
Scaffold	Full model	0.7615 ± 0.0308	–
	No cross-attention	0.7529 ± 0.0154	0.70
	No ranking loss	0.7658 ± 0.0302	0.87
	No diffusion head	0.7691 ± 0.0243	0.75
	Backbone only	0.7669 ± 0.0196	0.81

SD denotes standard deviation. p is a two-sided Welch two-sample test of each variant against the full configuration on the same split. At $n = 3$ the panel cannot separate differences of this size, so these rows bound what the design can show rather than attributing accuracy to any component.

C. EFFICIENCY AND SCALABILITY

We measured the cost of retaining the interaction map from the prediction's forward pass. All timings are single-process CPU numbers on an 11th-generation Intel Core machine using 8 threads, the relevant setting for a screening deployment without accelerators; the checkpoint itself was trained on GPU.

1) Where the parameters are

The model has 41.5 M parameters (Fig. 2a): cross-attention fusion 19.7 M (47.4%), protein adapter 13.3 M (32.1%), affinity head 5.8 M (13.9%), ligand graph encoder 2.0 M (4.9%), and diffusion head 0.70 M (1.7%). The interaction map is therefore not free in parameters, since fusion is the single largest component. What it buys is that those parameters are spent on the computation producing the prediction, not on a parallel explanation pathway.

2) Measured map-collection overhead

Table 4 compares per-pair latency with and without collecting the attention maps at each batch size. Both paths compute the attention weights; this comparison measures their collection overhead. The difference averages +1.5%, ranging from -1.4% to $+6.6\%$. These descriptive timings suggest small collection overhead; repeated timing runs would be needed to estimate its uncertainty.

3) Batch size and target length

Throughput is highest at one pair per batch, 204 ms per pair, and falls monotonically as the batch grows: 217 ms at 4 pairs, 232 ms at 8, 295 ms at 16, and 459 ms at 32 (Fig. 2b). At 64 pairs the run exhausts memory, requesting a single 34.9 GB allocation on a 64 GB machine.

Padding can contribute because the protein adapter's self-attention scales with batch size and the square of the longest sequence length. However, the loader sorts by length and excludes two warmup batches per setting, leaving different timed subsets (254 to 192 pairs). This sweep therefore does not

TABLE 4. CPU inference latency per pair, with and without collecting the interaction maps.

Pairs per batch	Scores only (ms/pair)	Maps collected (ms/pair)	Difference
1	204.0	209.6	+2.7%
4	216.6	213.6	−1.4%
8	232.3	233.7	+0.6%
16	295.5	292.6	−1.0%
32	458.9	489.2	+6.6%
64	out of memory (34.9 GB requested)		

isolate batch size from target composition. Within the single-pair run, a single-pair regression over targets from 292 to 4,128 residues gives 0.477 ms per residue with Pearson $r = 0.919$ ($n = 254$), while ligand size shows little linear association with latency ($r = -0.03$).

These measurements motivate testing small, length-grouped batches and choosing batch size for the target lengths and available memory.

4) Cached embeddings make screening practical

The timing sample used cached ESM-C embeddings for 132 unique targets, and the protein-embedding file occupies 385 MB; the combined graph and protein cache occupies 509 MB. Reusing these embeddings avoids running the protein language model for each candidate. The reported 210 ms per pair excludes initial embedding computation.

D. WHAT THE INTERACTION MAPS ENCODE

Fig. 3 shows what one of these maps looks like, and the rest of this section measures what such a map does and does not carry.

Contact benchmarks scored against a crystal pose cannot assess compounds without reference poses. The in-domain EGFR panel includes decoys that are presumed inactive without experimental confirmation. The target is fixed, the ligand varies over 300 actives and 3,000 property-matched decoys, and the model emits a map for every one. We extracted the per-residue profile for all 3,300 candidates; this rerun reproduced the released affinity scores to a correlation of 1.000000.

1) Most of the map is shared across the tested ligands

Decomposing the variance of the 3,300 by 1,210 profile matrix separates the panel-mean residue profile from the deviations specific to each tested ligand. The residue main effect accounts for 92.7% of the variance and the ligand-by-residue interaction for 7.3%; the ligand main effect is zero by construction, because each profile is a distribution over residues (Fig. 5a). Profiles correlate with the panel mean at 0.968 on average and with each other at 0.938 (Fig. 5b).

The practical reading is that an interaction map shown for a single compound is mostly a target-associated residue profile. Presenting one as an explanation specific to that compound overstates what distinguishes it from any other ligand, a caution that applies to attention heatmaps generally [33]–[35].

2) The remaining 7.3% carries signal

The residual profiles retain predictive information. Features are standardized per residue, so the shared residue component is removed before fitting and what follows uses only the ligand-specific part.

Logistic probes used shuffled stratified five-fold cross-validation with fixed L2 regularization ($C = 0.1$), balanced class weights, and standardization fitted within each training fold. Ridge regression used shuffled five-fold outer cross-validation, selecting among penalties 1, 10, 100, and 1,000 through five-fold inner cross-validation restricted to each outer training fold. Ridge regression from the profile to predicted affinity reaches a cross-validated R^2 of 0.321, against -0.029 for a shuffled target. A linear readout of the map therefore recovers part of the model’s predicted affinity.

The active versus decoy comparison is shown in Fig. 5c. A cross-validated logistic probe on the profile separates actives from decoys at AUROC 0.984, against 0.909 for a probe on ten bulk physicochemical descriptors and 0.499 under shuffled labels. A profile might merely encode ligand size, so we residualized it on those descriptors by fitting the descriptor-to-profile regression within each outer training fold and applying it to held-out candidates. The resulting AUROC is 0.963, suggesting label-associated information beyond this linear descriptor adjustment. These random candidate folds do not test generalization to new chemical series or targets. The analysis uses a fixed checkpoint and one set of folds; the shuffled controls use one fixed permutation.

One comparison must not be drawn from these numbers. The probes are supervised and fitted on this panel, whereas the model’s own affinity score (AUROC 0.839) was not fitted on panel labels. The fingerprint baseline (0.992) uses known active references without fitting a classifier. These different uses of labels prevent a direct performance comparison. The probes suggest that the maps contain information associated with active–decoy labels, subject to DUD-E’s decoy-selection bias.

3) Agreement with crystallographic contacts

Residue-level attention agrees with crystallographic contacts to varying degrees across the five complexes (Fig. 4). ABL1 with imatinib reaches AUROC 0.709 (95% CI 0.591 to 0.827) and CDK2–3QS 0.702 (0.568 to 0.837), both excluding chance, whereas CDK2–LS2, EGFR, and FAK1 do not. The panel is strongly heterogeneous ($Q = 15.5$ on 4 df, $I^2 = 74\%$), so the random-effects estimate of 0.585 (0.470 to 0.701, $p = 0.15$) is the defensible summary, not the fixed-effect 0.586.

This small panel does not establish a shared localization effect or explain the observed variation. A larger panel with verified structural annotations is needed.

E. RETRIEVAL VALIDATION

Library prioritization is the application most often claimed for a drug–target affinity model, so it deserves the strictest test. We evaluate it only on the two panels whose positives are

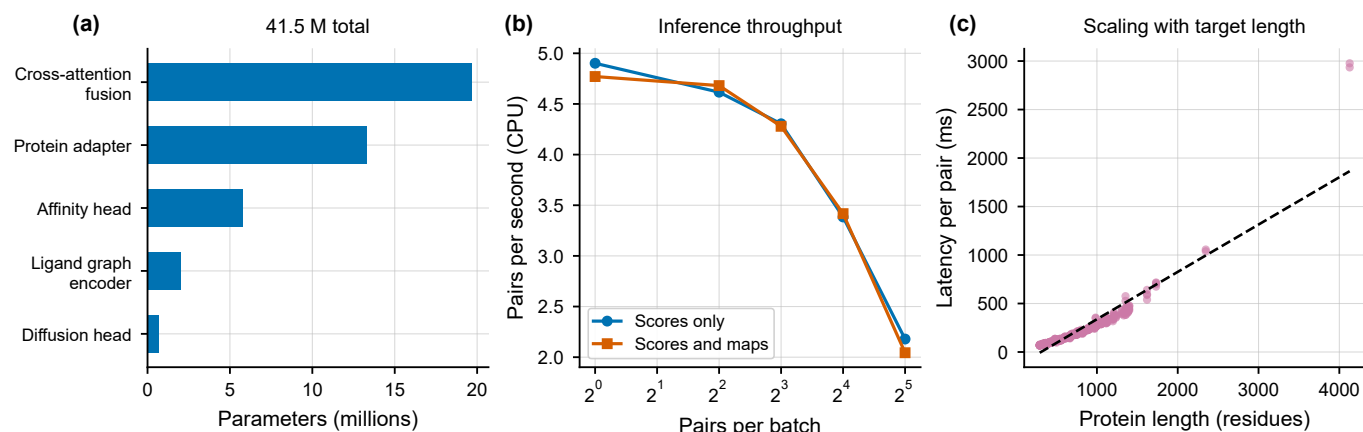


FIGURE 2. Cost profile of the inference path, measured on CPU. (a) Parameters by module; the cross-attention fusion that produces the interaction maps is the largest single component at 47.4% of 41.5 M parameters. (b) Throughput against pairs per batch, with and without collecting the attention maps. The curves are similar, with an average latency difference of 1.5%. Timed subsets differ across batch sizes because two warmup batches are excluded from each length-sorted run. (c) Single-pair latency against target length, a major determinant of time and memory.

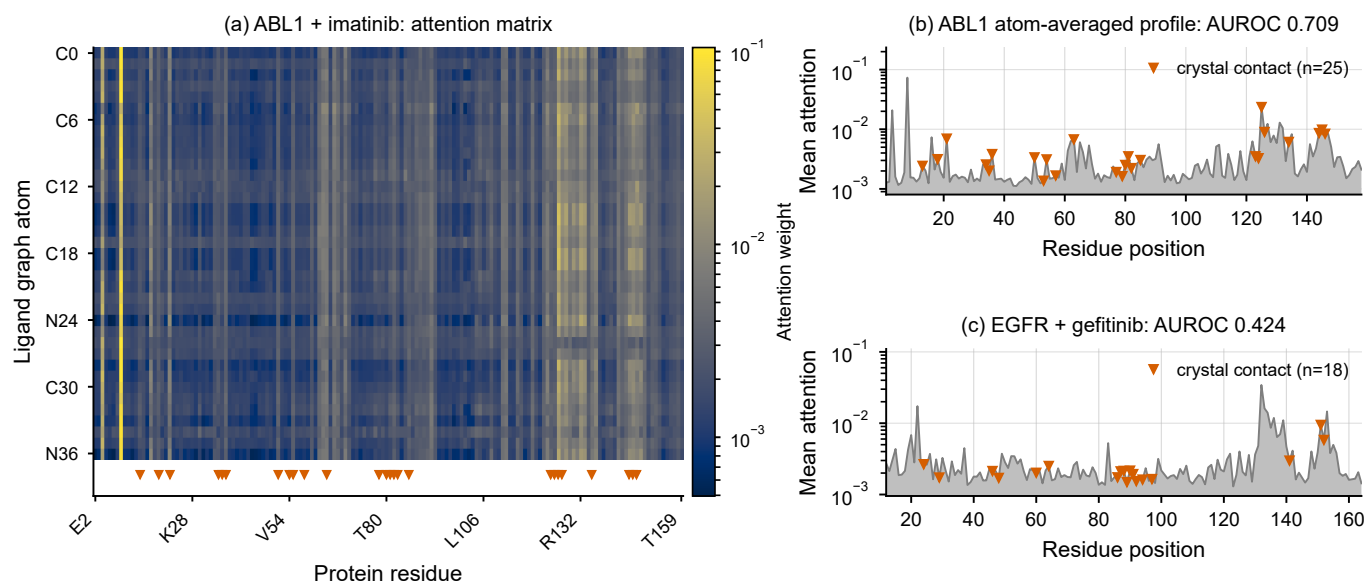


FIGURE 3. Attention maps and their agreement with crystallographic contacts. (a) The atom-by-residue attention matrix for ABL1 with imatinib (PDB 2HYI), taken from the same forward pass that produces the affinity prediction. Rows are ligand graph atoms, columns residues, and the triangles below mark the 25 residues within 4.5 Å of the ligand in the crystal. Color is on a log scale because the weights span more than two decades. The pattern is almost entirely vertical, meaning every ligand atom attends to nearly the same residues, and the brightest bands fall in the contact-rich region. (b) The corresponding released per-residue profile, obtained by averaging attention over atoms, at a residue contact AUROC of 0.709. (c) The per-residue profile for EGFR with gefitinib, whose interval includes chance. Attention localizes in two of the five complexes (Fig. 4), so (a) is the better case rather than the typical one. Panels show contact-region windows; AUROCs use all evaluated residues. The matrix and released profiles come from separate inference runs.

assay-defined, using a ligand-similarity baseline on both and AutoDock Vina on EGFR.

1) Out of domain

The similarity baseline scores each candidate by its fingerprint resemblance to the nearest known active, with that candidate itself excluded when it is one of them. On the H1 panel this baseline reached AUROC 0.996 and AUPRC 0.806, recovering 17 of 20 actives within the top 1% and all 20 within the top 5%, while DeepDTA-iBAM reached 0.667 ± 0.008 and 0.021 ± 0.018 (mean $\pm$ SD across five libraries), recovering 14% within the top 10%. H1 is a G-protein-coupled receptor

and the model was trained on kinase-dominated data, so this measures transfer rather than performance on the model's own family. The small replicate variance shows the weak signal is reproducible across background libraries rather than an artifact of one draw.

2) In domain, against similarity and docking

The in-domain EGFR panel provides a complementary test: assay-defined actives, property-matched decoys, and a target from the model's own family. Across all 3,300 candidates the fingerprint baseline reached AUROC 0.992 and recovered 97.0% of actives within the top 10%, against 0.839 and 60.7%

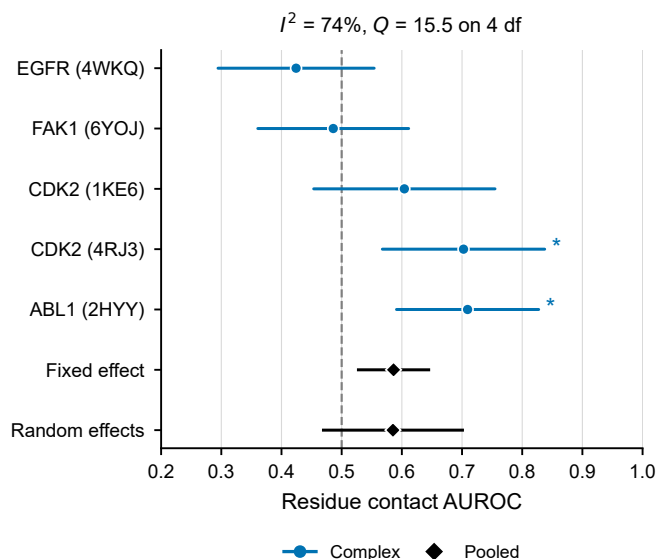


FIGURE 4. Residue contact AUROC by complex, with analytic Hanley–McNeil 95% confidence intervals computed from the 250 to 300 residues within each complex rather than treating each complex as one observation. Asterisks mark intervals excluding chance. Diamonds give fixed-effect and random-effects pooled estimates. The wider random-effects interval reflects between-complex heterogeneity; individual intervals are unadjusted for multiplicity.

for the model, a paired model-minus-fingerprint difference of -0.153 (95% CI -0.184 to -0.125).

Docking places that gap on an external scale (Table 5, Fig. 6a). On the docked set of all 300 actives and 460 decoys, all of which prepared successfully, Vina reached AUROC 0.648, the model 0.836, and the fingerprint baseline 0.992. Both paired bootstraps resolve: the model leads Vina by 0.188 AUROC (95% CI $+0.139$ to $+0.237$) and trails the fingerprint baseline by 0.156 ($+0.125$ to $+0.189$), the latter reproducing the full-panel ordering and giving a similar margin.

These comparisons have distinct limitations. Receptor preparation used Open Babel, and Vina used one rigid receptor conformation. The gap behind the fingerprint baseline holds on two target panels, but DUD-E requires decoys topologically dissimilar to its actives, which favors fingerprint methods [17], [19], [20]. The 0.156 gap may therefore overstate the difference on a panel without this selection bias.

Similarity search provides a strong reference on these panels when known actives are available. The learned scorer warrants further evaluation when reference actives are unavailable or when interaction maps and a shared latent representation are also needed.

F. LOCAL DESIGN

Under EGFR conditioning from dasatinib the diffusion head returned 100 unique valid molecules within 4,000 decode attempts, against 100 for fragment swaps and 40 for random atom edits, so it has a yield advantage over random edits only.

Diffusion proposals have a median fingerprint similarity of 0.14 to the seed, against 0.59 for fragment swaps and 0.84 for random edits (Fig. 7a), giving them the lowest median

similarity among the three generators. Fragment swaps score better on all four reported chemical heuristics (Fig. 7b): quantitative estimate of drug-likeness (QED, 0.562 versus 0.402), synthetic accessibility (2.93 versus 4.03), Lipinski pass rate (0.94 versus 0.44), and alert-free rate (0.74 versus 0.38) [36]–[39].

Diffusion did give the highest mean predicted affinity, 11.289 against 11.196 for random edits and 11.079 for fragment swaps, with the returned seed at 11.269 in the same candidate-scoring pass. That cannot establish superiority: every candidate is rescored by the checkpoint that produced the diffusion proposals, so it measures agreement between a generator and its own critic, and both margins are below the held-out RMSE.

Auditing the set molecule by molecule found two problems the aggregate statistics hid; the full audit is in the supplement. One of the 100 molecules is dasatinib itself, so the seed was retained in its own output. Another 39 of the remaining 99 carry motifs that pass RDKit sanitization but are not credible follow-up candidates under the audit criteria, leaving 60 analogs.

Figure 8 shows the seed and the five highest-scoring analogs among these 60, ordered by their saved predictions. Predicted KIBA scores span 11.855 to 11.993, compared with 11.269 for the seed in the same scoring pass. All remain below the conventional KIBA activity threshold of 12.1. Four fail the reported Lipinski screen, and the highest-scoring analog also triggers a PAINS/BRENK alert. These examples illustrate why model ranking must be considered alongside chemical filters. Potential stereocenters are unspecified in their SMILES; the saved docking scores refer to the prepared structures and do not compare all possible stereoisomers.

The second problem is the more consequential (Fig. 7c). Uniqueness had been assessed by canonical SMILES, and on that criterion all 100 molecules are distinct, with 98 distinct Bemis–Murcko scaffolds, the ring systems and the linkers between them. That reads as near-complete scaffold diversity. Yet all 100 share a *single* generic framework, the same skeleton with every atom type erased, and it is dasatinib’s [40]. This follows mechanically from a decoder that holds bond topology fixed, so it is not a malfunction, but the head is enumerating substitutions around one skeleton rather than exploring chemical space. Diversity should be reported by generic framework, not by counting SMILES.

Docking supplies the missing scorer (Fig. 6b). Of those 60 analogs 56 could be prepared; 4 failed charge assignment, indicating remaining preparation limitations. The 56 scored a mean of -8.364 kcal/mol against -7.646 for the seed, with 55 of 56 better than it.

Read this narrowly. Vina scores track molecular size and these are substitutions on the seed’s own skeleton, so a systematic shift is unsurprising. The result provides a comparison independent of the model’s affinity head, with lower scores than the starting structure for 55 analogs. It does not establish improved potency or synthetic feasibility.

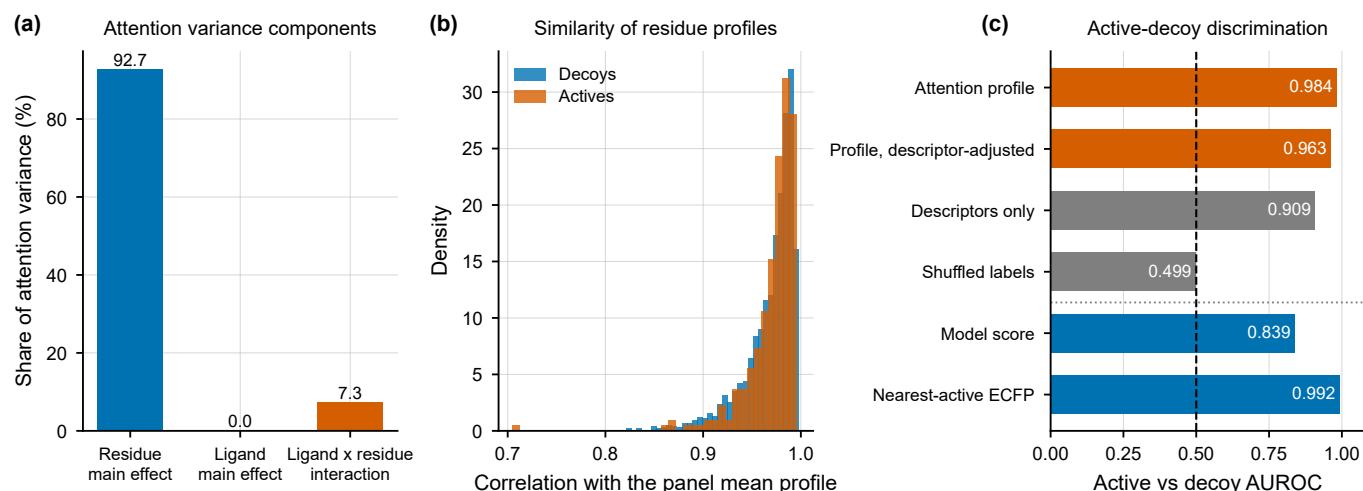


FIGURE 5. What the interaction maps encode, measured on 300 assay-defined EGFR actives and 3,000 property-matched decoys against a fixed target. (a) Variance decomposition of the ligand-by-residue attention matrix: 92.7% of the variation is the residue main effect shared across the tested ligands, and 7.3% is the ligand-specific interaction. (b) Each ligand's profile correlates with the panel mean profile at 0.968 on average, so a map shown for one compound is largely target-associated. (c) Active versus decoy AUROC. The first four bars are supervised probes fitted by cross-validation; the last two are rankers without panel-specific classifier fitting and are not directly comparable to the probes; the fingerprint ranker uses known active references. A probe on the profile beats a probe on bulk descriptors, and still does so after the profile is residualized on those descriptors. Standardization and descriptor residualization were fitted within each training fold; estimates use combined out-of-fold predictions under random candidate splits.

TABLE 5. In-domain EGFR retrieval on the docked set: 300 assay-defined actives and 460 property-matched decoys.

Ranker	AUROC	AUPRC	BEDROC20	EF 1%	EF 5%	Rec. 10%
AutoDock Vina	0.648	0.493	0.463	0.95	1.07	0.129
DeepDTA-iBAM	0.836	0.836	0.976	2.53	2.47	0.250
Nearest-active ECFP	0.992	0.992	1.000	2.53	2.53	0.253

Higher is better throughout. Vina reports an estimated binding free energy, so its score is negated before ranking. AUROC is insensitive to prevalence alone, but all metrics may change with candidate composition. AUPRC and enrichment factors also depend on prevalence and are reported at the realized 39.5% active fraction, so they cannot be compared with enrichment figures quoted for the full 1:10 panel. EF denotes enrichment factor at the indicated ranked fraction; Rec. 10% is the fraction of actives recovered in the top 10%. EF, recovery, and BEDROC average over all orderings within tied-score groups.

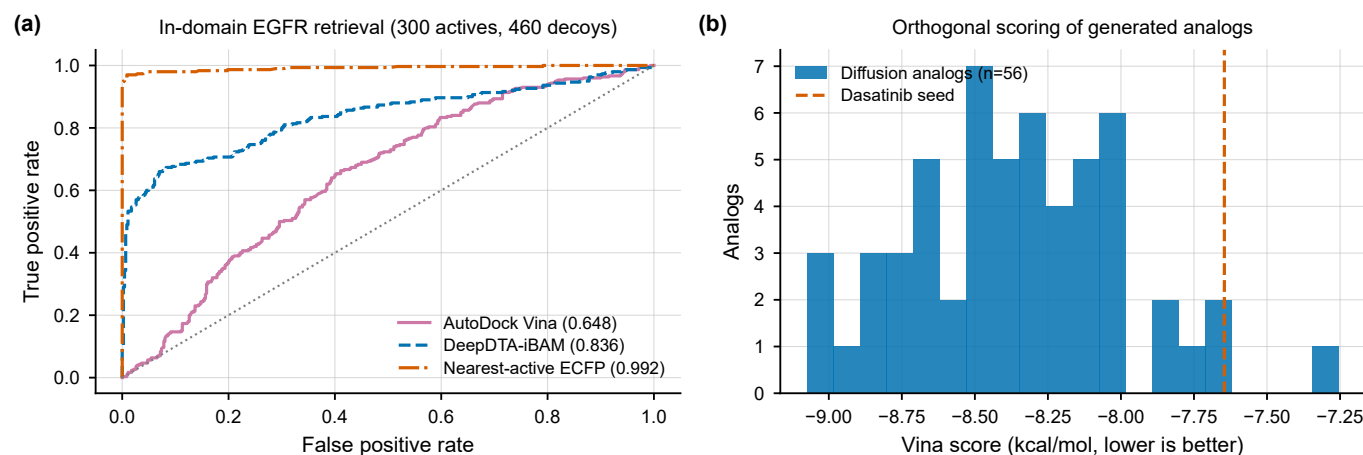


FIGURE 6. An external structure-based reference for both secondary capabilities. (a) Retrieval on the in-domain EGFR panel. The learned scorer outranks AutoDock Vina, and a nearest-active fingerprint search outranks both. Vina used Open Babel receptor preparation and a single rigid conformation, which bounds the first comparison. Panel (a) shows ROC curves for 300 actives and 460 decoys. (b) Vina scores for the 56 audited diffusion analogs that could be prepared, against the dasatinib seed. Lower is better. The analogs include 55 with lower scores than the seed under a scorer that played no part in generating them.

V. EVALUATION METHODOLOGY

Building the panels ourselves exposed two constructions that prevent valid interpretation. The remedy depends on the failure:

a single-class AUROC needs different labels, while matching on similarity requires overlapping class supports.

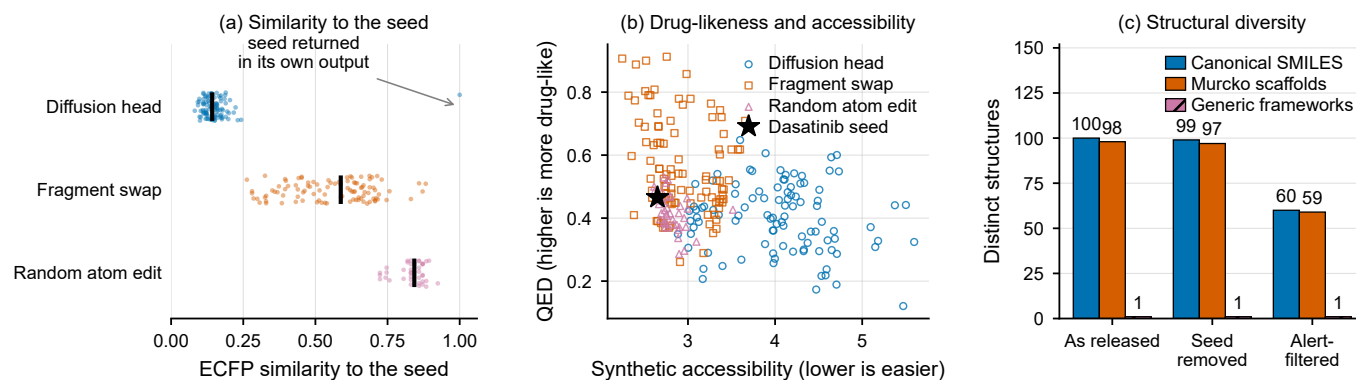


FIGURE 7. What the analog-generation head does, on 100 diffusion proposals, 100 fragment swaps, and 40 random atom edits conditioned on EGFR from a dasatinib seed. (a) Fingerprint similarity of each proposal to the seed, with the median marked. Diffusion proposals have the lowest median similarity; the single proposal at 1.0 is the seed, returned in its own output. (b) Drug-likeness against synthetic accessibility for the same molecules. Diffusion proposals have lower QED and higher synthetic accessibility scores than fragment swaps on average. (c) Distinct structures through the audit under three definitions. Canonical SMILES and Bemis–Murcko scaffolds track the set size, which reads as near-complete diversity, but generic frameworks, the same skeletons with atom types and bond orders erased, stay at one throughout. The head varies substituents widely around a single fixed skeleton.

A. BASE-RATE SATURATION

In the evaluated kinase complexes essentially every ligand heavy atom lies within 4.5 Å of protein. Across our five complexes the atom contact counts were 26 of 27, 37 of 37, 29 of 29, 31 of 31, and 34 of 34: every ligand atom was a contact in four of the five, and 96.3% in the fifth (Fig. 9a).

These contact labels describe the reference structures and do not depend on model predictions. At the chosen 4.5 Å cutoff, the atom-level labels are saturated in this panel.

Two consequences follow mechanically. Contact AUROC is undefined when every label is the same, so whatever a results table prints there is a software artifact. And top- k overlap, where k is the number of true contacts, has a chance expectation of exactly k/n , the contact base rate: at a base rate of 0.993 the observed 0.992 is an excess over chance of -0.0003 (95% CI -0.001 to $+0.001$). Without the base rate beside it, the raw overlap reads as strong attribution and a fallback AUROC as inverted attribution, although these summaries provide little evidence of attribution.

The residue-level half remains well posed, with contact base rates of 5.7% to 9.5%, which is exactly the contrast the base rate makes visible. At the atom level, informative evaluation requires labels or metrics that distinguish atoms, such as graded buried surface area or a panel with more solvent-exposed ligand atoms.

B. SELECTION-RULE CIRCULARITY

The EGFR family panel of Section III-C required Tanimoto similarity ≥ 0.40 to gefitinib or erlotinib, and the anchors came from that same filtered family, so all 347 molecules satisfy the constraint by construction.

The consequence is stronger than leakage. Positives span 0.400 to 0.904 in nearest-anchor similarity and decoys span 0.054 to 0.233, so the largest decoy value falls 0.167 units below the smallest positive and not one of the 2,341 candidates lies in that interval (Fig. 9b). Any threshold inside the gap separates the classes perfectly, which is why a nearest-anchor

fingerprint ranker reaches AUROC 1.000 with a bootstrap interval of zero width. That number describes the selection rule, not the baseline.

Matching or reweighting on similarity requires overlap in the class-conditional similarity ranges. Because this panel has none, these approaches cannot balance similarity without changing the candidate set or label definition. Comparing the two ranges identifies this limitation before attempting such adjustments.

The in-domain panel behaves differently: 97 of its 300 actives and 2,950 of its 3,000 decoys fall in a shared similarity region. This permits similarity matching but does not remove DUD-E’s bias toward fingerprint methods.

C. THREE CHECKS

Both failures are detectable from output a pipeline already produces.

- 1) **Report the positive-class base rate** beside every enrichment metric, and report an undefined AUROC as undefined rather than substituting a fallback value.
- 2) **Reference top- k overlap to chance**, as excess over the base-rate expectation, because raw overlap approaches 1.0 automatically as the base rate does.
- 3) **Audit baseline independence from label construction**. Check whether positive or decoy selection uses the scored similarity function. Disclose any resulting bias when interpreting that baseline.

An audit of 25 published studies indicates the first is not yet routine: of the 11 reporting a quantitative attention-versus-residue metric, only 3 report the base rate beside it. The circular construction did not appear in that sample, so we diagnose it without claiming it common.

VI. DISCUSSION

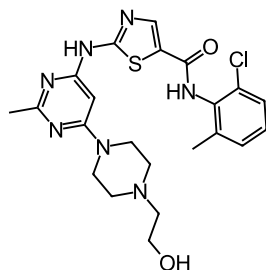
A. WHAT THE SYSTEM SUPPORTS

Auditing the secondary capabilities as strictly as the primary one leaves a narrower but clearer recommendation.

Dasatinib and the five highest-scoring audited analogs

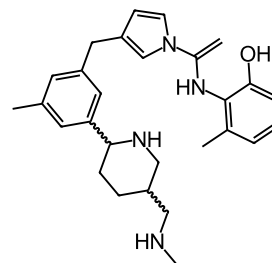
Ranked by saved model-predicted EGFR KIBA score among 60 audit survivors

(a) Dasatinib seed



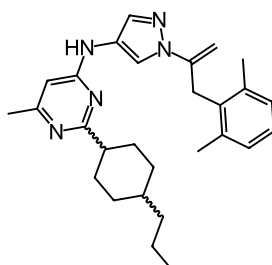
Predicted KIBA 11.269 | Δ KIBA +0.000
 ECFP Tanimoto 1.000 | QED 0.466 | SA 2.65
 Vina -7.646 kcal/mol
 Lipinski: pass | PAINS/BRENK: pass

(b) Rank 1: analog_000



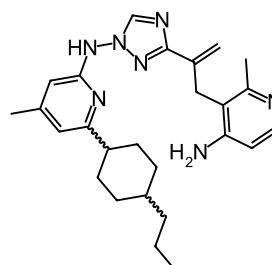
Predicted KIBA 11.993 | Δ KIBA +0.723
 ECFP Tanimoto 0.162 | QED 0.361 | SA 3.97
 Vina -8.430 kcal/mol
 Lipinski: fail | PAINS/BRENK: fail

(c) Rank 2: analog_001



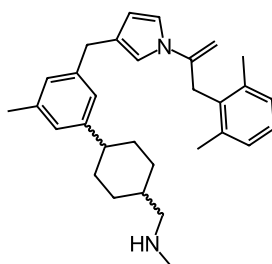
Predicted KIBA 11.918 | Δ KIBA +0.648
 ECFP Tanimoto 0.204 | QED 0.402 | SA 3.10
 Vina -8.346 kcal/mol
 Lipinski: fail | PAINS/BRENK: pass

(d) Rank 3: analog_002



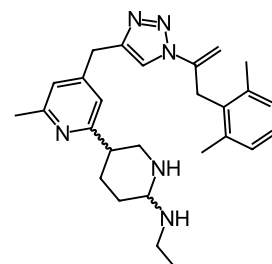
Predicted KIBA 11.905 | Δ KIBA +0.636
 ECFP Tanimoto 0.130 | QED 0.477 | SA 3.54
 Vina -8.310 kcal/mol
 Lipinski: fail | PAINS/BRENK: pass

(e) Rank 4: analog_003



Predicted KIBA 11.896 | Δ KIBA +0.626
 ECFP Tanimoto 0.143 | QED 0.391 | SA 3.03
 Vina -8.839 kcal/mol
 Lipinski: fail | PAINS/BRENK: pass

(f) Rank 5: analog_004



Predicted KIBA 11.855 | Δ KIBA +0.586
 ECFP Tanimoto 0.150 | QED 0.542 | SA 4.22
 Vina -8.463 kcal/mol
 Lipinski: pass | PAINS/BRENK: pass

FIGURE 8. Dasatinib seed and the five highest-ranked EGFR-conditioned diffusion analogs among 60 nonseed candidates retained by the four-motif audit. Ranking uses saved predicted KIBA scores; Δ KIBA is the difference from the seed scored in the same pass. ECFP Tanimoto compares radius-2, 2,048-bit fingerprints with the seed. Higher QED indicates greater drug-likeness; lower synthetic accessibility (SA) scores indicate easier predicted synthesis. Vina values are archived docking scores, with lower values preferred. Lipinski and PAINS/BRENK screening flags are separate from the four-motif audit. Wavy bonds indicate unspecified stereochemistry. Predictions, heuristics, and docking scores do not establish potency or synthetic feasibility.

Supported: ranking within the evaluated chemical space. Pooled concordance index 0.8625 and AUROC 0.9240 support prioritization on the tested KIBA pairs, but do not establish ranking accuracy within individual target series. Limits of agreement span -0.895 to 0.876 KIBA units, limiting single-compound interpretation. The split contains no unseen

compound or target, so these results describe interpolation within known chemical space.

Supported: interaction maps as a target-associated readout. The map comes from the same forward pass as the prediction and carries ligand-specific information a linear probe recovers, but the shared residue profile accounts for

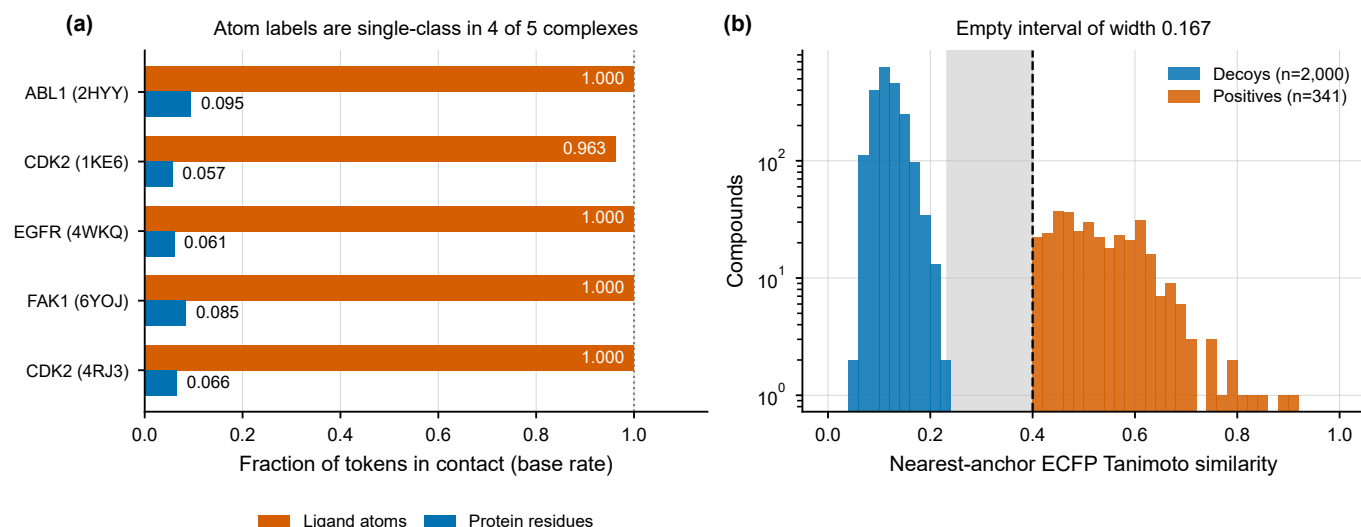


FIGURE 9. Two benchmark constructions that prevent valid interpretation. (a) Contact base rates in the five-complex kinase panel. Essentially every ligand heavy atom is in contact, leaving a single-class label vector in four of five complexes, while the residue-level base rate of 5.7% to 9.5% leaves a well-posed problem. (b) Class-conditional nearest-anchor similarity for the EGFR family panel. Positives were selected by requiring similarity ≥ 0.40 to the reference binders from which the anchors were later drawn (dashed line); decoys were sampled with no similarity constraint. The supports are disjoint, separated by a shaded interval of width 0.167 containing no molecule of either class, so no similarity-matched subpanel can be drawn and the comparison cannot be repaired by reweighting.

92.7% of its variance on the EGFR panel. Contact localization remains uncertain at the panel level.

Potential use: screening without a reference active set.

The model outranks AutoDock Vina by 0.188 AUROC and scores a candidate in 210 ms against a median of 104 s per ligand for docking, on a workstation without an accelerator. Where a target has a structure but no known actives, this motivates evaluation of a learned scorer; performance on previously unseen targets remains unestablished.

Not supported: prioritizing a library in preference to a similarity search. On the evaluated panels, a fingerprint search using known actives leads the model by 0.156 AUROC in domain and also performs better out of domain.

Partially supported: proposing follow-up chemistry. The diffusion head produces the lowest median fingerprint similarity to the seed among the three generators, and 55 of its 56 audited proposals have lower docking scores than the seed. However, it has no yield advantage over fragment swaps, fewer than half its molecules pass the Lipinski rules against 94% for fragment swaps, every proposal shares the seed's generic framework, and 4 of 60 could not be prepared for docking. As a substitution enumerator it works; as a source of chemical novelty it is unevidenced.

B. LIMITATIONS

The structural panel has five complexes across four targets, including two CDK2 complexes, which limits target-level independence. Its variance components put the requirement at 18 to 21 complexes to separate a residue contact AUROC of 0.585 from chance at 80% power, against an estimated 0.20 to 0.30 under the assumed effect and variance. These estimates assume independent complexes and do not include uncertainty

in the effect size or variance components.

Except for the archived three-seed ablations, analyses use one checkpoint and one training seed. The component ablation cannot be strengthened without retraining at a scale we did not run.

The retrieval comparisons are limited differently. H1 is out of domain, so it tests transfer, and DUD-E's decoy selection may favor the fingerprint baseline in domain, as discussed in Section IV-E. The docking reference used Open Babel receptor preparation and a single rigid conformation, so it is a reference point rather than an optimized campaign.

Finally, both failure modes were characterized in panels we built, which is what lets each be traced to a construction step. The literature audit extends the first reporting gap to published work, without establishing saturation in those studies; the prevalence of the second failure remains open.

VII. CONCLUSION

We presented DeepDTA-iBAM, a multi-task architecture in which bidirectional cross-attention between ligand atoms and protein residues supports affinity prediction, interaction-map extraction, and retrieval. Analog generation shares the protein adapter and checkpoint. The design claim is an interface claim rather than an accuracy claim, and we measured it: the map is a byproduct of the forward pass that yields the prediction. Retaining it had a measured mean collection overhead of 1.5%.

Auditing the secondary capabilities as strictly as the primary one is the second half of the contribution. EGFR maps share a dominant residue profile but carry a ligand-specific component from which a linear readout predicts part of the predicted affinity and distinguishes active-decoy labels, including for compounds with no crystal pose. Retraining ablations cannot single out any one component, though disabling the fusion

pathway increases the trained model's RMSE by 33%, and at retrieval the model outranks an established docking program but trails a nearest-active fingerprint search.

Two benchmark constructions examined here prevent valid interpretation, and three reporting checks detect both with minimal additional computation, separating a benchmark that can be repaired from one that must be rebuilt. The path forward is a matched follow-up in one target family: assay-defined positives with property-matched decoys and no topological constraint [20], [21], a larger co-crystal panel with verified structural annotations, and orthogonal scoring of generated compounds [41], [42].

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Supplementary Material

for

DeepDTA-iBAM: Cross-Attention Interaction Mapping for Affinity Prediction, Ligand Retrieval, and Analog Generation

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This supplementary information extends the main manuscript with uncertainty estimates, panel compositions, derivations, and compound-level outputs. These materials document where the reported conclusions are stable, where they remain sensitive to evaluation design, and how the underlying benchmark outputs were organized. All sections, subsections, figures, and tables use canonical supplementary numbering with the S prefix.

Four findings from the main text carry into this file and should be kept in view when reading it. First, atom-level contact metrics in the structural localization benchmark are saturated, because the atom contact base rate is 96.3 to 100%. Atom contact AUROC is undefined in 4 of 5 complexes; the residue-level metrics remain valid and are analyzed in the main text by inverse-variance pooling. Second, the EGFR retrieval panel defined its positive class using a fingerprint similarity cutoff to the reference binders from which the anchors were drawn. The class-conditional similarity distributions are entirely disjoint, positives spanning 0.400 to 0.904 and decoys 0.054 to 0.233, so fingerprint baselines are not valid comparators on that panel, no similarity-matched subpanel can be constructed from these candidates, and the latent and affinity results are also conditional on this selection rule. Third, head-to-head comparisons between rankers scored on a shared candidate set require a paired test; marginal intervals reported in the tables below overlap in cases where the paired difference is nevertheless decisive. Fourth, the set of 100 diffusion analogs contains the dasatinib seed itself and 39 structures carrying implausible motifs, and its 100 canonically unique SMILES collapse to a single Bemis–Murcko generic framework; the audited set is reported in the main text.

The two assay-defined retrieval comparisons, the out-of-domain H1 panel and the in-domain EGFR panel with assay-defined actives and property-matched decoys, are reported in the main text; the construction of the latter is documented in Section S7. Section S2.1 documents the circular panel’s composition in full, Section S3.1 gives the label counts behind the saturation result, Section S5 derives the power calculation quoted in the main text, and Section S6 reports an audit of reporting practice in 25 published studies, so every diagnosis can be checked directly against the released data.

S1 Additional Benchmark Context

The following benchmark material expands the evaluation on the custom random 90/5/5 KIBA split (termed standard in the main text) with uncertainty and scaffold-aware context, clarifying how stable the main benchmark estimates are and how they should be interpreted relative to more chemically stringent settings.

S1.1 Residual Diagnostics

The main text reports pooled ranking performance while limiting single-compound interpretation because of the width of the limits of agreement. Fig. S1 shows a small overall mean bias of -0.010 , although residual medians vary modestly across the measured range. Dispersion remains substantial: the 95% limits of agreement span -0.895 to 0.876 KIBA units, showing substantial individual prediction error despite the small mean bias. These pooled diagnostics do not establish accuracy within individual target series.

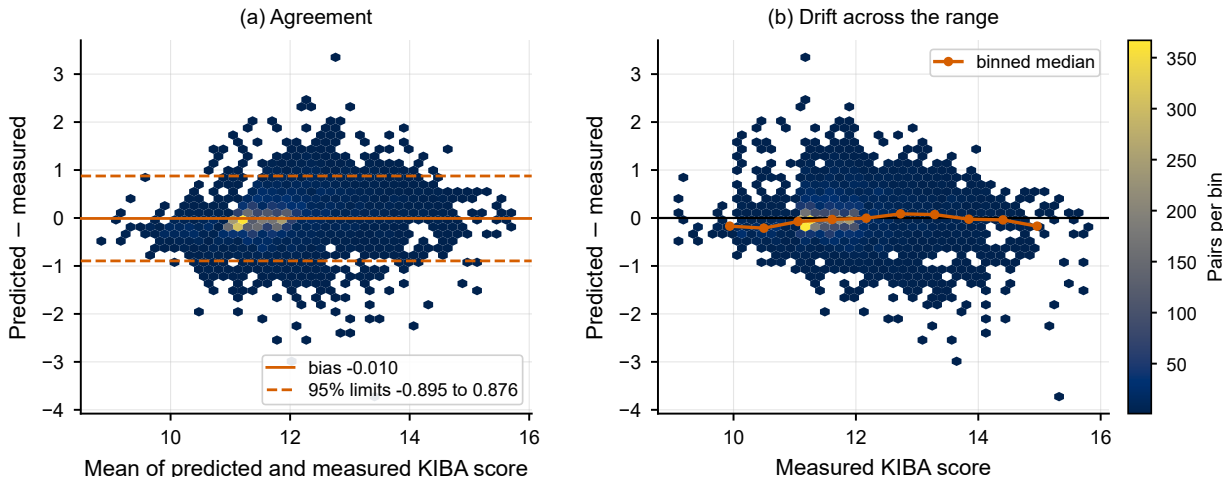


Figure S1: Residual diagnostics for the standard-split KIBA evaluation ($n = 5,913$ held-out pairs). (a) Bland–Altman agreement between predicted and measured KIBA score, with the mean bias and the 95% limits of agreement. (b) Residual against the measured value, with medians for bins containing at least 20 observations among 12 equal-width bins. Both panels use hexagonal binning, and the shared color scale gives the count per bin. The overall bias is small relative to the dispersion, which limits single-compound use.

S1.2 Bootstrap Uncertainty

The bootstrap table below complements the primary benchmark values by showing local uncertainty around the reported point estimates. These intervals describe evaluation uncertainty for the saved checkpoint, but they should not be mistaken for between-study uncertainty because the literature benchmark rows were not rerun in a shared environment. Two concordance index values are computed from the same saved predictions using different conventions for ties. The main text reports 0.8625, from the released `analysis/analysis_benchmark_metrics.py`, which excludes pairs with equal measured values and counts pairs with equal predictions as half a concordance, consistent with the comparison methods. Table S1 reports 0.8621, the earlier in-house point estimate that scored prediction ties as zero, accompanied by a

600-resample bootstrap interval. The archived ablation summary differs for a separate reason: it averages three independently trained seeds rather than this single saved rerun.

Table S1: Bootstrap uncertainty for the single-checkpoint standard-split evaluation.

Metric	Estimate	95% CI lower	95% CI upper
RMSE	0.4517	0.4365	0.4672
MAE	0.2950	0.2869	0.3046
Concordance	0.8621	0.8556	0.8676
Pearson r	0.8741	0.8634	0.8840
R^2	0.7049	0.6795	0.7269

Metrics were recomputed over 600 bootstrap resamples of the held-out prediction pairs ($n = 5,913$) to obtain nonparametric 95% confidence intervals. RMSE and MAE are root-mean-square error and mean absolute error in KIBA units; CI denotes confidence interval; r is Pearson correlation; R^2 is the coefficient of determination. These intervals quantify local evaluation uncertainty. The custom random KIBA split differs from published benchmark folds, so between-paper comparisons remain contextual.

S1.3 Scaffold-Split Ablation Context

The archived scaffold-split ablation summary evaluates model variants under a more chemically stringent split. Differences in mean concordance are modest relative to variability across three retraining seeds, limiting conclusions about the contribution of each component.

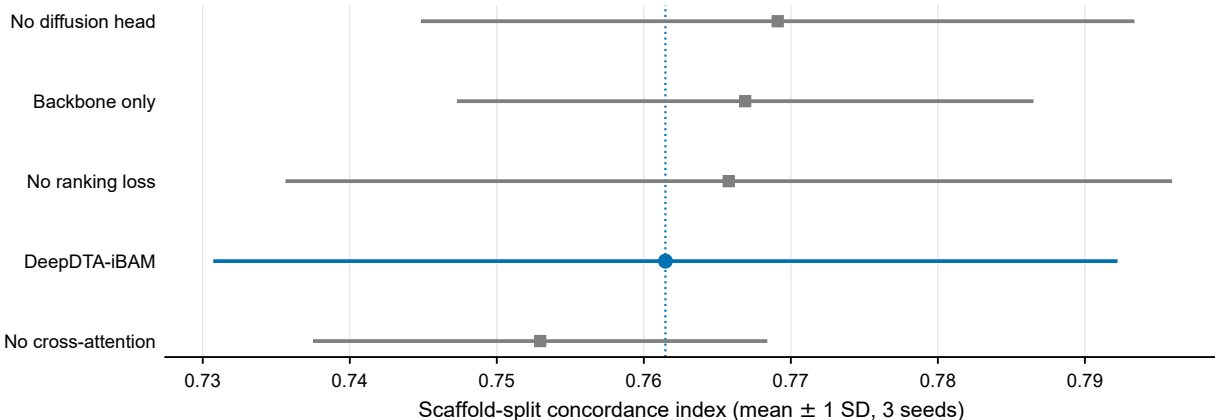


Figure S2: Scaffold-split ablation summary from the archived three-seed aggregate results. Markers give the mean scaffold-split concordance index across three retraining seeds and bars span one standard deviation, for the full model (blue circle, dotted reference line) and each ablated variant. A higher concordance index means better ranking agreement under chemistry shift. The means differ modestly relative to between-seed variability; these bars do not constitute a formal paired comparison. The no-diffusion-head and backbone-only variants have slightly higher means, and no-cross-attention the lowest. The main text reports a separate inference-time ablation of the fusion pathway.

S2 Retrieval Analyses

These materials document the composition of the circular EGFR panel and its full metric set. ECFP rows in those tables are leakage diagnostics rather than comparators, and the remaining results are conditional

on the same selection rule. H1 replicate values are available in `results/h1_drug_fishing_replicate_metrics.csv`.

S2.1 EGFR Panel Composition and the Fingerprint-Circularity Diagnosis

The EGFR family was assembled from ChEMBL target CHEMBL203 by filtering exact-binding records to molecules with $\text{pChEMBL} \geq 7.0$ and Morgan/ECFP Tanimoto similarity ≥ 0.40 to gefitinib (CHEMBL939) or erlotinib (CHEMBL553). All 347 retained molecules satisfy the similarity criterion by construction, with an observed minimum of exactly 0.4000, a mean of 0.5011, and a maximum of 1.0000. The 6 interpolation anchors were then drawn from within this same family, and the 2,000 ZINC decoys were sampled with no similarity constraint.

A nearest-anchor ECFP ranker evaluated on this panel is thus scored on the criterion that defined the positive class. The observed consequence is perfect separation with a degenerate bootstrap interval (AUROC = 1.000 with 95% CI [1.000, 1.000], AUPRC = 1.000, and BEDROC20 = 1.000), consistent with selection-rule circularity. Model-derived rankings remain mutually comparable on this selected panel, but their absolute performance is conditioned on its similarity-based composition. Section S7 documents the assay-defined replacement panel.

S2.2 Expanded EGFR Retrieval Outputs

The following tables provide the full retrieval metric set, including the exploratory combined score, together with the top exploratory ZINC candidates. The saved circular panel used 6 anchors, 341 holdouts, 2,000 local ZINC decoys, and a 500-resample candidate-set bootstrap.

Table S2: Full EGFR retrieval benchmark, including the exploratory combined path-plus-affinity score.

Method	AUROC	AUPRC	BEDROC20	EF at 1%	EF at 5%	Recovery at 5%	Recovery at 10%
Interpolation path	0.8152	0.5378	0.6774	6.8651	5.5270	0.2786	0.3812
Latent nearest anchor	0.7958	0.5092	0.6522	6.8651	5.2943	0.2669	0.3519
Predicted affinity	0.9495	0.9250	0.9757	6.8651	6.8651	0.3460	0.6774
Nearest-anchor ECFP	1.0000	1.0000	1.0000	6.8651	6.8651	0.3460	0.6891
Anchor-centroid ECFP	1.0000	1.0000	1.0000	6.8651	6.8651	0.3460	0.6891
Combined (exploratory)	0.9484	0.8574	0.9337	6.8651	6.8651	0.3460	0.6540

The circular panel contains 341 holdouts and 2,000 decoys (positive fraction, 0.146). AUROC and AUPRC denote areas under the receiver operating characteristic and precision-recall curves; BEDROC20 is Boltzmann-enhanced discrimination of ROC with $\alpha = 20$. EF denotes enrichment factor; recovery is the fraction of holdouts retrieved at the stated ranked fraction. The combined score reuses model-derived information in latent geometry and affinity scoring and is included because it ranked the exploratory ZINC hits. ECFP denotes extended-connectivity fingerprint; its rows are leakage diagnostics, not comparators.

S3 Interpretability Outputs

The per-complex table below resolves the heterogeneity behind the structural-localization summary and makes the base-rate degeneracy directly inspectable.

Table S3: Top 10 exploratory ZINC candidates ranked by the combined interpolation and affinity score.

Rank	ZINC ID	Nearest anchor	Tanimoto	Affinity	Path distance	QED	SA	Lipinski	Alert-free
1	ZINCi500000c8vtF	CHEMBL5271803	0.1207	11.8189	0.4453	0.8071	3.7965	1	0
2	ZINCj6000002mmyC	GEFITINIB	0.1264	11.7613	0.4978	0.7847	2.8978	1	1
3	ZINCf6000000b7au	CHEMBL6054692	0.1310	12.0211	0.5600	0.6743	2.5243	1	1
4	ZINCf7000000bqB0	POZIOTINIB	0.0805	12.1240	0.5909	0.6545	3.2409	1	0
5	ZINCi60000010vpq	POZIOTINIB	0.1477	11.8373	0.5472	0.7538	2.8602	1	1
6	ZINCk5000003VAyo	GEFITINIB	0.1667	11.7046	0.5208	0.5806	2.9466	1	1
7	ZINCk5000003VM6m	CHEMBL5271803	0.1416	11.7866	0.5436	0.7121	3.4476	1	1
8	ZINCj8000002pP9B	GEFITINIB	0.1364	11.2123	0.4464	0.7970	2.9873	1	1
9	ZINCf8000000bTgg	CHEMBL6054692	0.1348	11.7523	0.5637	0.7167	3.9291	1	1
10	ZINCf8000000bQkW	GEFITINIB	0.1375	11.6596	0.5494	0.6060	3.6083	1	0

Predicted EGFR affinity is in KIBA units; path distance is measured in the model’s latent space. Tanimoto denotes nearest-anchor ECFP similarity. QED is quantitative estimate of drug-likeness (higher is more favorable); SA is synthetic accessibility score (lower indicates easier synthesis). Lipinski and alert-free columns report pass status (1, yes; 0, no). Full SMILES strings are retained in the CSV artifact. These entries are exploratory follow-up candidates, not experimentally validated EGFR actives.

S3.1 Contact Base Rates Underlying the Degeneracy Diagnosis

Table S4 reports the label composition for each complex. Residue-level label vectors contain both classes (5.7–9.5% positives), whereas atom-level label vectors are single-class in 4 of 5 complexes and contain exactly one negative in the fifth. Atom contact AUROC is therefore undefined in 4 of 5 complexes, and atom top- k overlap, where k is the number of true contacts, has a base-rate expectation of 0.993, against which the observed 0.992 represents an enrichment of 1.00 $\times$.

Table S4: Label composition of the structural-localization benchmark.

PDB	Protein	Residues	Contact res.	Res. rate	Atoms	Contact atoms	Atom rate
1KE6	CDK2	280	16	0.057	27	26	0.963
2HYY	ABL1	263	25	0.095	37	37	1.000
4RJ3	CDK2	287	19	0.066	29	29	1.000
4WKQ	EGFR	296	18	0.061	31	31	1.000
6YOJ	FAK1	259	22	0.085	34	34	1.000

For each complex, the number of residues and ligand heavy atoms is reported alongside the number and fraction annotated as crystallographic contacts at a 4.5 Å heavy-atom cutoff. The atom-level contact fraction reaches 1.000 in 4 of 5 complexes, which renders atom contact AUROC undefined and saturates atom top- k overlap.

Table S5 gives the per-complex localization scores and perturbation results underlying the main-text summaries.

S4 Local Design Outputs

The remaining local-design materials provide compound-level context for the EGFR-conditioned analog-proposal benchmark. These outputs remain model-internal; the rule-based chemistry filters reported here are triage cues, not orthogonal validation of potency or developability.

Table S5: Per-complex structural localization benchmark across 5 kinase co-crystal complexes.

PDB	Protein	Ligand	Residue AUROC	Atom AUROC	Residue top- <i>k</i>	Atom top- <i>k</i>	Residue mask	Atom mask
1KE6	CDK2	LS2	0.6042	0.2692	0.0625	0.9615	-0.0882	0.5457
2HYY	ABL1	STI	0.7092	NA	0.2000	1.0000	-0.0065	-0.1940
4RJ3	CDK2	3QS	0.7025	NA	0.1053	1.0000	0.0829	-1.5489
4WKQ	EGFR	IRE	0.4243	NA	0.0556	1.0000	0.3978	1.1355
6YOJ	FAK1	P4N	0.4860	NA	0.1364	1.0000	-0.9607	-0.3156

Each row reports residue-contact and atom-contact AUROC, top-*k* overlap with *k* equal to the number of true contacts, and perturbation signals for one protein–ligand pair. NA indicates undefined atom-contact AUROC for single-class labels; see Table S4. Masking signals are differences in predicted-affinity reduction (KIBA units) between masking the top 10 tokens and masking 10 randomly selected tokens. Positive values favor the attention-selected set. The random control used one draw per complex, so uncertainty across random draws was not estimated. Atom-contact labels were mapped from PDB order to the cached graph order by enumerating all element- and bond-preserving mappings. All mappings give the same contact vector for each complex. A targeted CPU rerun corrects the 1KE6 atom AUROC to 0.2692; the other atom AUROCs remain undefined and all atom overlaps are unchanged. Residue metrics and masking results retain their archived values. The audit is released as `analysis/revalidate_atom_contacts.py`.

S4.1 Structural Audit of the Generated Set

The main text summarizes the audit of the 100 released diffusion analogs; Table S6 gives it in full. The seed row is dasatinib itself, recovered at Tanimoto 1.000 and counted among the generated analogs. Structural alerts identify motifs excluded by the audit despite passing RDKit sanitization; molecules may carry more than one, and the final column gives how many of each class were nonetheless passed as alert-free by the PAINS/BRENK filter used in the original pipeline. Generic frameworks are Bemis–Murcko scaffolds with atom types and bond orders erased: the set collapses to one, so canonical-SMILES uniqueness does not measure scaffold diversity here.

Table S6: Structural audit of the 100 released diffusion analogs.

Stage	Molecules	Unique SMILES	Murcko scaffolds	Generic frameworks
As released	100	100	98	1
Seed removed	99	99	97	1
Alert-filtered	60	60	59	1
<i>Structural alerts removed (molecules may carry more than one)</i>			<i>Passed PAINS/BRENK</i>	
Peroxide O–O	11			0
Low-valent P	4			0
Ring hydrazine N–N	17			7
Ring hydroxylamine N–O	18			7
Any alert	39			13

The upper block follows the set through the audit, from the released output, through removal of the dasatinib seed, to the 60 analogs surviving the structural-alert filter, counting distinct canonical SMILES, distinct Bemis–Murcko scaffolds and distinct generic frameworks at each stage. Canonical-SMILES and scaffold counts track the set size while the generic-framework count stays at one. The lower block breaks the removed molecules down by alert class, with the number of each class that the original PAINS/BRENK filter had nonetheless passed as alert-free.

S4.2 Ranked Diffusion Analog Table

SMILES and properties for all 60 analogs surviving the audit are in the released file `results/generated_egfr_analogs_audited.csv`. Table S7 prioritizes chemical screening flags and QED. The structure figure in the main text ranks this same pool by predicted KIBA score alone, so the two selections serve different purposes.

Table S7: Top 10 diffusion analogs ranked by chemical filters and drug-likeness.

#	Murcko framework (rings)	MW (Da)	cLogP	QED	SA	Tanimoto	Lip.	Alert-free
1	4 rings (3 aromatic)	445.6	3.96	0.550	3.74	0.123	yes	yes
2	4 rings (3 aromatic)	444.6	4.31	0.542	4.22	0.150	yes	yes
3	4 rings (3 aromatic)	444.6	4.36	0.540	3.34	0.140	yes	yes
4	4 rings (3 aromatic)	444.6	4.49	0.537	4.12	0.149	yes	yes
5	4 rings (2 aromatic)	452.6	1.53	0.523	4.10	0.157	yes	yes
6	4 rings (3 aromatic)	444.6	4.74	0.468	4.24	0.134	yes	yes
7	4 rings (3 aromatic)	445.6	4.94	0.457	4.04	0.127	yes	yes
8	4 rings (3 aromatic)	444.6	4.30	0.508	3.08	0.223	yes	no
9	4 rings (3 aromatic)	444.6	3.81	0.461	4.15	0.161	yes	no
10	4 rings (3 aromatic)	450.5	4.30	0.440	4.11	0.158	yes	no

Drawn from the 60 molecules surviving removal of the dasatinib seed and of the 39 structures carrying peroxide, low-valent phosphorus, or cyclic hydrazine and hydroxylamine motifs. Rows are ordered by Lipinski pass status, alert-free status, descending QED, ascending synthetic accessibility, and descending predicted EGFR affinity. All entries have four rings and share one Bemis–Murcko generic framework with dasatinib; aromatic ring counts can differ. MW is molecular weight in Da; cLogP is calculated octanol–water partition coefficient on a logarithmic scale. QED is quantitative estimate of drug-likeness; SA is synthetic accessibility score. Lip. denotes Lipinski pass status, and alert-free denotes passing the original PAINS/BRENK filter. Tanimoto is similarity to the dasatinib starting structure. These are model-internal follow-up candidates, not validated optimization outcomes.

S5 Power Calculation for the Structural-Localization Panel

The main text gives a conditional estimate of 18 to 21 co-crystal complexes to distinguish a residue contact AUROC of 0.585 from chance at 80% power. This section gives the derivation and its assumptions. The calculation is implemented in `analysis_power.py`.

S5.1 Design and Variance Components

Complexes are the unit of replication. The five complexes represent four targets, including two CDK2 complexes, which limits target-level independence. Each complex j supplies one residue contact AUROC A_j estimated from its own 250 to 300 residues, so A_j carries a within-complex sampling variance s_j^2 on top of a between-complex variance τ^2 . The within-complex variance is the analytic Hanley–McNeil quantity [1],

$$s_j^2 = \frac{A_j(1 - A_j) + (n_j^+ - 1)(Q_1 - A_j^2) + (n_j^- - 1)(Q_2 - A_j^2)}{n_j^+ n_j^-},$$

with $Q_1 = A_j/(2 - A_j)$ and $Q_2 = 2A_j^2/(1 + A_j)$, and n_j^+ , n_j^- the contact and non-contact residue counts. The between-complex variance is the DerSimonian–Laird estimator [2] applied to the same five complexes. On the observed panel these evaluate to

$$\bar{s}^2 = 0.00455 \quad (\bar{s} = 0.067), \quad \tau^2 = 0.01284 \quad (\tau = 0.113),$$

so between-complex heterogeneity contributes roughly three times the variance that within-complex sampling error does. This is the quantitative form of the $I^2 = 74\%$ reported in the main text, and it is why adding residues to existing complexes cannot substitute for adding complexes.

S5.2 Required Panel Size

Assuming independent complexes, a random-effects pooled estimate over k complexes has

$$\text{Var}(\hat{A}_{\text{pooled}}) \approx \frac{\tau^2 + \bar{s}^2}{k} = \frac{0.01739}{k},$$

so testing $H_0 : A = 0.5$ against $A_1 = 0.585$ at two-sided α with power $1 - \beta$ requires

$$k \geq \frac{(z_{1-\alpha/2} + z_{1-\beta})^2(\tau^2 + \bar{s}^2)}{(A_1 - 0.5)^2} = \frac{(1.960 + 0.842)^2 \times 0.01739}{0.085^2} = 18.9,$$

that is 19 complexes. Substituting the empirical complex-level variance (0.01631, without separating variance components) gives 18. Repeating the calculation with a one-sample t test and power computed from the noncentral t distribution gives 20 to 21. The resulting estimate is **18 to 21 complexes**. This range reflects the variance estimate and small-sample approximation; it does not include uncertainty in the assumed effect or variance components, which were estimated from only five complexes. Table S8 gives the full curve.

The estimated power of the 5-complex panel under these assumptions is 0.30 under the normal approximation and 0.20 under the t calculation. Thus, a nonsignificant result would be likely even if the assumed effect were present; these conditional calculations do not establish whether that effect is real.

S6 Audit of Reporting Practice in the Published Literature

The reporting checks proposed in the main text are only worth proposing if they are not already standard. To test that, we audited 25 published drug–target affinity and compound–protein interaction studies for whether they apply them. The audit is implemented in `analysis_literature_audit.py` and its verdicts are released in `literature_audit.csv`.

S6.1 Scope and Criteria

Papers were drawn from Europe PMC and restricted to open-access full texts, so that every verdict can be checked by a reader against the same source we read. The sample spans 2018 to 2026 and mixes the canonical architectures (DeepDTA, DeepAffinity, MolTrans, MGraphDTA) with recent work through 2026. It is a

Table S8: Statistical power by number of protein–ligand complexes.

Complexes k	SE of pooled AUROC	95% CI half-width	Power (normal)	Power (t)
5	0.0590	0.1156	0.30	0.20
10	0.0417	0.0817	0.53	0.44
15	0.0340	0.0667	0.70	0.64
20	0.0295	0.0578	0.82	0.78
25	0.0264	0.0517	0.90	0.87
30	0.0241	0.0472	0.94	0.93
40	0.0208	0.0409	0.98	0.98
50	0.0186	0.0365	1.00	0.99

Variance components from the observed panel: $\tau^2 = 0.01284$, $\bar{s}^2 = 0.00455$
 Estimated for 80% power at $\alpha = 0.05$: $k = 18$ to 21 complexes

Standard errors and power are conditional on variance components estimated from the observed 5-complex panel, $\tau^2 = 0.01284$ and $\bar{s}^2 = 0.00455$, and a target residue contact AUROC of 0.585 against chance at two-sided $\alpha = 0.05$. CI half-widths use the normal approximation. The normal power column uses the large-sample approximation; the t column uses a one-sample t test and the noncentral t distribution. Neither calculation incorporates uncertainty in the assumed effect or variance components.

convenience sample of the open-access literature rather than a systematic review, and it is not a random draw from all published models; papers behind subscription paywalls, including several widely cited ones, could not be included on the same terms and were excluded rather than assessed from their abstracts.

Each paper was read for four questions. Does it validate attention or importance scores against annotated binding-site or contact residues, in any form, including a purely visual case study? If so, is that validation quantitative, meaning it reports an overlap, enrichment, accuracy, intersection-over-union or AUROC number, rather than a picture with residues named in the caption? If quantitative, does the paper anywhere report the positive-class base rate, the fraction of residues that are pocket or contact residues, alongside the metric? And separately, does it define a positive or active set using a chemical similarity threshold or analog series while also scoring a fingerprint or similarity baseline on that same set? A paper that computes a base rate internally but does not print it is recorded as not reporting it, because the check we propose is a reporting practice rather than a computation.

S6.2 Result

Sixteen of the 25 papers validate attention against annotated residues. Five of those 16 do so only visually, showing a heatmap or naming agreeing residues in a caption without any metric, which leaves nothing to check a base rate against. Eleven report a quantitative figure, and **3 of those 11 report the positive-class base rate beside it**. The remaining 8 report an enrichment, overlap, or accuracy number with no statement of what fraction of residues were positives, which is the condition under which base-rate saturation is invisible. The three exceptions (GGAP-CPI, CDPN, and LiBr) show that the practice is achievable and is occasionally followed; two of the three are recent studies whose explicit purpose was benchmark quality.

The second criterion returned nothing. **None of the 25 papers scored a similarity baseline on a similarity-defined positive set**, so selection-rule circularity was not observed anywhere in this sample. The reason is structural rather than reassuring: most drug–target affinity papers do not build target-family retrieval

panels at all, and those in this sample that ran virtual screening evaluations drew their actives from annotation databases such as DUD-E and LIT-PCBA rather than from a similarity cutoff. We therefore make no claim about how often selection-rule circularity occurs in published work. Eight of the eleven quantitative studies omit the base rate needed to assess saturation; the audit does not establish that their metrics were saturated. Selection-rule circularity was diagnosed in our own panel, and its prevalence in published work remains unknown.

Table S9: Audit of 25 published drug–target affinity and compound–protein interaction studies.

Model / study	Venue	Year	Residue claim	Quantitative	Base rate
DeepDTA	Bioinformatics	2018	no	–	–
DeepAffinity	Bioinformatics	2019	yes	yes	no
MolTrans	Bioinformatics	2021	yes	no	–
MGraphDTA	Chem Sci	2022	yes	yes	no
AttentionSiteDTI	Brief Bioinform	2022	yes	no	–
PIGNet	Chem Sci	2022	no	–	–
GraphATT-DTA	Biomedicines	2023	yes	no	–
PocketDTA	J Cheminform	2024	yes	no	–
CASTER-DTA	Brief Bioinform	2025	yes	yes	no
MIF-DTI	Brief Bioinform	2025	no	–	–
GGAP-CPI	J Chem Inf Model	2025	yes	yes	yes
CDPN	J Chem Inf Model	2025	yes	yes	yes
GPS-DTI	BMC Biol	2025	yes	yes	no
GenSPARC	Commun Chem	2025	yes	yes	no
HMM-DTA	J Cheminform	2026	no	–	–
ChemGLaM	J Cheminform	2026	yes	yes	no
DCI-SiteDTA	BMC Bioinformatics	2026	yes	yes	no
DTAP	Brief Bioinform	2026	yes	no	–
TS EDTA	Bioinformatics	2026	yes	yes	no
DCGAT-DTI	Bioinform Adv	2026	no	–	–
TriDTI	Brief Bioinform	2026	no	–	–
MMCA	PLoS One	2026	no	–	–
Dual-encoder fusion	BMC Bioinformatics	2026	no	–	–
LiBRe	J Chem Inf Model	2026	yes	yes	yes
PhenoModel	Acta Pharm Sin B	2026	no	–	–
16 of 25 validate attention against annotated residues; 11 of those do so quantitatively					
Of the 11 quantitative evaluations, 3 report the positive-class base rate and 8 do not					
0 of 25 score a similarity baseline on a similarity-defined positive set					

Residue claim is whether the paper validates attention or importance against annotated binding-site or contact residues in any form. *Quantitative* distinguishes a reported metric from a visual case study. *Base rate* is whether the fraction of residues that are pocket or contact residues is reported alongside that metric, and is not applicable where the validation is purely visual. All entries are open-access full texts, listed with their PMC identifiers in the released CSV so that each verdict can be checked.

S7 Construction of the In-Domain, Assay-Defined EGFR Panel

The main text reports an in-domain retrieval comparison that replaces the circular EGFR panel. This section documents how it was built, including one route that failed, because the failure is instructive about why property-matched decoys are hard to produce from a general-purpose compound library.

S7.1 Why the Decoys Are Not Drawn From a General-Purpose Library

The saved exploratory EGFR candidate file contains 6,060 unique molecule IDs and SMILES, each with $\text{pChEMBL} \geq 7.0$ and a recorded measurement type of K_i , IC_{50} , or K_d . This file and the consolidation cache are distinct artifacts. The latter contains 5,976 potent molecule IDs, which consolidate to 5,942 parent IDs. All 5,976 are present in the exploratory file, together with 84 additional molecules: 73 are absent from the consolidation cache and 11 have only subthreshold potency in that cache. These counts explain the numerical discrepancy; the historical query or database-version cause of the different records cannot be established from the retained files.

Recomputing descriptors from the saved SMILES gives mean molecular weight 506.0 Da and cLogP 4.77 for the 6,060 candidates, versus 253.6 Da and 0.01 for the 60,000-compound ZINC pool. The pool spans 199.2 to 438.3 Da. Only 8 candidates fall within all six marginal descriptor ranges for molecular weight, cLogP, hydrogen-bond donors and acceptors, rotatable bonds, and formal charge. This range check does not establish that adequate matched neighbors exist. The saved matching output contains one active and ten decoys, but its exact historical matching criteria could not be reconstructed. These property differences support using the DUD-E EGFR panel for the reported comparison; the balance of that panel is quantified below. The reproducible audit is released as `analysis/revalidate_egfr_curation.py`.

S7.2 The Panel Used

DUD-E’s EGFR target supplies actives defined by ChEMBL activity with decoys property-matched to them by construction [3]. From its 542 actives and 35,050 decoys we sampled 300 and 3,000, respectively, under a fixed seed. Property balance between the classes, expressed as a standardized mean difference, is 0.06 for molecular weight, -0.06 for hydrogen-bond donors, 0.10 for rotatable bonds, 0.76 for hydrogen-bond acceptors, 0.88 for cLogP, and -0.91 for net formal charge. The first three are well matched; the last three retain substantial imbalance.

The class-conditional similarity diagnostic uses the maximum ECFP Tanimoto similarity to gefitinib or erlotinib, as for the circular panel. Here the supports *overlap*: actives span 0.058 to 1.000 and decoys span 0.021 to 0.214 , and the shared region contains 97 of the 300 actives and 2,950 of the 3,000 decoys. This overlap permits construction of a similarity-matched subpanel, although it does not establish an unbiased benchmark. The retrieval fingerprint baseline uses a separate definition: nearest-active similarity to all 300 panel actives, with each active excluded from its own reference set.

S7.3 What This Panel Still Does Not Settle

DUD-E selects decoys that are topologically dissimilar to its actives. This does not define the positive class by similarity, but it can favor fingerprint methods. Published critiques show that models can perform well on such benchmarks by exploiting dataset biases [4, 5]. The 0.153 AUROC gap between the fingerprint baseline and the model may therefore overstate the difference on a panel without topological decoy selection. Evaluating that difference requires an independent panel with experimentally supported inactive compounds, such as a suitable target-specific subset of LIT-PCBA [6], and remains open.

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